

When should I suspect a hyperglycaemic emergency?

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The hyperglycaemic complications of [diabetic ketoacidosis \(/topics/diabetes-type-2/background-information/complications/\)](#) (DKA) and [hyperosmolar hyperglycaemic state \(/topics/diabetes-type-2/background-information/complications/\)](#) (HHS) can be life-threatening and require emergency hospital [admission \(/topics/diabetes-type-2/management/management-adults/#managing-suspected-hyperglycaemic-emergencies-dka-hhs\)](#) if suspected.

- Suspect a diagnosis of DKA in a person with known diabetes or significant hyperglycaemia (blood glucose level greater than 11 mmol/L) *and* the following:
 - Clinical symptoms
 - Polydipsia and polyuria.
 - Weight loss.
 - Abdominal pain, nausea and/or vomiting.
 - Shortness of breath.
 - Lethargy, drowsiness, and/or confusion.
 - A possible history of [precipitating factor\(s\) \(/topics/diabetes-type-2/diagnosis/when-to-suspect-hyperglycaemic-emergencies-dka-hhs/#precipitating-factors\)](#).
 - Clinical signs
 - Fruity smell of acetone on the breath.
 - Tachypnoea, acidotic breathing (deep sighing 'Kussmaul respiration').
 - Tachycardia, dehydration (may present with reduced skin turgor, sunken eyes, prolonged capillary refill time) or shock (tachycardia, hypotension, drowsiness, reduced urine output).
 - Blood or urinary ketones
 - In an adult, test for blood or urinary ketones, even if plasma glucose levels are near normal.
 - In a child or young person, test for blood ketones, even if plasma glucose levels are near normal. If this is not possible, arrange immediate admission to a hospital with acute paediatric facilities.
 - Note: ketones are high if urinary ketones are greater than 2+, or capillary blood ketones are above 3 mmol/L. Test for blood ketones using a ketone testing meter and strips. See the CKS topic on [Insulin therapy in type 2 diabetes \(/topics/insulin-therapy-in-type-2-diabetes/\)](#) for information on ketone testing meters and strips.
 - Note: urinary ketones must be read 15 seconds after the stick is dipped.
- Consider the possibility of DKA in all people with type 2 diabetes who are unwell, even with normal blood glucose levels.
 - Children and young people on insulin therapy may develop DKA with normal blood glucose levels, as may people using [SGLT-2 inhibitors \(/topics/diabetes-type-2/prescribing-information/sglt-2-inhibitors/\)](#), and people with a history of alcohol excess and/or chronic liver disease.
 - Low blood ketone levels (less than 3 mmol/L) do not always exclude DKA.

- **Suspect a diagnosis of HHS if a person is unwell with severe hyperglycaemia (blood glucose level typically above 30 mmol/L) for several days and:**
 - Clinical symptoms
 - Disorientation, confusion, and/or drowsiness.
 - Polyuria and polydipsia.
 - Nausea.
 - A possible history of [precipitating factor\(s\)](#) ([/topics/diabetes-type-2/diagnosis/when-to-suspect-hyperglycaemic-emergencies-dka-hhs/#precipitating-factors](#)).
 - Clinical signs
 - Severe dehydration and hypovolaemia.
 - No significant signs of ketosis, or blood or urinary ketones on testing.

Precipitating factors

- **Precipitating factors of diabetic ketoacidosis (DKA) and hyperosmolar hyperglycaemic state (HHS) include:**
 - Infection.
 - Inadequate insulin or non-adherence with insulin treatment.
 - New onset of diabetes mellitus or other physiological stress (such as trauma or surgery).
 - Other medical conditions (such as hypothyroidism or pancreatitis).
 - Drugs (such as corticosteroids, diuretics, atypical antipsychotics, and sympathomimetic drugs such as salbutamol).

[[MHRA, 2016](#) ([/topics/diabetes-type-2/references/](#)); [Wolfsdorf, 2018](#) ([/topics/diabetes-type-2/references/](#)); [French, 2019](#) ([/topics/diabetes-type-2/references/](#)); [MHRA, 2020](#) ([/topics/diabetes-type-2/references/](#)); [NICE, 2020b](#) ([/topics/diabetes-type-2/references/](#))]

Basis for recommendation

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